

元智大學
YUAN ZE UNIVERSITY

管理學院財務金融暨會計碩士班
財務金融碩士學程

碩士論文計畫書
MASTER'S THESIS PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形楔子
**The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's
Platform Economy**

研究生 / Student: 王新福 Christopher Ongko
指導教授 / Advisor: 孔德蓉 De-Rong Kong

中華民國 115 年 9 月
September 2026

Abstract

This study measures the gap between the transaction value digital platforms process in Indonesia and the revenue those platforms recognize for themselves. I call that gap the invisible wedge and measure it with an Ecosystem Ratio, defined as unbooked gross transaction value divided by booked platform revenue. Because no Indonesian platform discloses a strictly country-labelled matched pair, every observation is admitted under a named evidence tier and tiers are never pooled. Seventeen platform-year levels across five issuer series spanning FY2019 to FY2025 show that the wedge moves, and disclosed revenue components explain the largest movements. Evidence from forty-eight matched issuer-years outside Indonesia indicates that the wedge is a general property of platform intermediation whose magnitude depends on business model. Official statistics and Indonesia's marketplace withholding rules then locate the activity and the records beyond platform accounts.

1. Introduction

1.1 Research Background and Motivation

Southeast Asia's digital economy grew from \$98 billion in 2020 to \$263 billion in 2024 (Google, Temasek and Bain 2024). Southeast Asia's digital economy grew from \$100 billion in gross merchandise value in 2020 to \$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). Digital platform scale, however, is not the same as tax-administrative visibility. These platforms record transactions, determine participant payouts, and book their own revenue, while the gross receipts earned by merchants, drivers, and other participants may remain outside routine third-party income reporting. I call this administrative information gap the "invisible wedge." The empirical question is whether this digitally recorded but separately reported flow is large enough to matter for tax administration.

the invisible wedge using an Indonesia-focused sample of Grab, GoTo, and Shopee, and then applies the resulting retained share estimates to calibrate four additional markets within the Association of Southeast Asian Nations (ASEAN): Vietnam, the Philippines, Thailand, and Malaysia. This paper evaluates the invisible wedge using an Indonesia-focused longitudinal sample of listed platform companies whose filings report both gross transaction value and platform revenue for the same period and the same business scope.

Grab, GoTo, and Shopee provide Listed Indonesian platform companies provide useful empirical settings because they directly intermediate transactions between consumers and merchants or service providers and publish financial disclosures from which transaction values and platform revenues can be observed or transparently derived. Their dual role as transaction intermediaries and record keepers places them at the center of the paper's third-party-reporting framework.

The term invisible refers to the administration's information channel, not to criminal concealment. Suppose a platform processes 100 units of transaction value but recognizes only 10 units as revenue. The remaining 90 units are still recorded within the platform's system but do not constitute platform revenue. Instead, they may represent merchant receipts, driver payouts, inventory costs, taxes paid elsewhere, and other pass-through payments. The paper measures this accounting gap without assuming that the residual represents profit, taxable income, or unpaid tax.

I measure the wedge with an ecosystem ratio defined as the gross transaction value (GTV) not recognized as platform revenue divided by the revenue that the platform does recognize. ~~The baseline estimate is constructed based on the disclosures in fiscal year of 2023 disclosures. Inputs for Grab and Shopee are either Indonesia-specific or transparently derived for Indonesia. By contrast, GoTo reports the required inputs directly only at the Group level, which includes operations outside Indonesia. Consequently, the estimated wedge of US\$66.8 billion should be interpreted as an Indonesia-focused baseline estimate. The corresponding figures for Vietnam, the Philippines, Thailand, and Malaysia are scenario-based calibrations derived from this benchmark rather than independently measured estimates. Because the relationship between transaction value and revenue also changes, each year-to-year transition is measured as the difference between transaction-value growth and revenue growth, and material divergences are reconciled against disclosed revenue components. For FY2023, the three leading Indonesian platform cases place transaction value not booked as platform revenue at approximately US\$40.07 billion, equivalent to roughly 2.9 percent of Indonesia's nominal gross domestic product in that year. That figure is a sum of three documented cases under mixed evidence classes rather than an Indonesia-wide total, and the longitudinal evidence then asks whether the gap is widening, narrowing, or reversing.~~

~~The paper measures gross participant-facing transaction value that is not booked as platform revenue; it does not estimate participant profit, unpaid tax, or value added.~~

The thesis therefore asks how large the invisible wedge is in Indonesia's platform economy, how it changes through time, what mechanisms explain those changes, and how much of the participant-facing activity already recorded by platforms enters routine third-party reporting and tax administration.

That motivating question is examined through the four linked research questions set out in Table 1. The first two establish the measurement and its movement; the third tests whether the pattern is specific to Indonesia; the fourth follows the activity and the records beyond platform accounts.

Table 1. Research questions and principal evidence. RQ1 and RQ2 concern the measurement and its movement within Indonesia; RQ3 tests generality outside Indonesia; RQ4 follows the activity and the underlying records beyond platform accounts.

	<u>Research question</u>	<u>Principal evidence</u>
<u>RQ1</u>	<u>How large is the invisible wedge in Indonesia, and does it persist across platforms and years?</u>	<u>17 admitted platform-year levels across three evidence tiers, FY2019–FY2025</u>
<u>RQ2</u>	<u>What explains movements in the wedge?</u>	<u>Disclosed revenue components: incentives, gross and net revenue, monetization, business scope</u>
<u>RQ3</u>	<u>Is the wedge specific to Indonesia or a general property of platform business models?</u>	<u>48 matched issuer-years across eight non-Indonesian platform businesses</u>
<u>RQ4</u>	<u>Where does activity outside platform revenue appear, and can the underlying records reach administration?</u>	<u>BPS national, channel and recordkeeping evidence; PMK 37/2025 and DJP materials</u>

1.1 Contributions1.2 Contributions

This study makes four contributions. First, it constructs a 2023 Indonesia-focused platform estimate and uses the combined 5.88% take rate only as a transparent calibration for four additional ASEAN markets. Derived and Group-level inputs are explicitly identified by provenance. This study makes four contributions. First, it measures the invisible wedge in Indonesia on a longitudinal basis. Earlier evidence for this market rests on a single fiscal year; the core sample assembled here observes transaction value and platform revenue for the same period and business scope across seventeen admitted platform-years spanning FY2019–FY2025, which makes it possible to report not only the scale of the wedge but its direction of travel.

Second, it explains why the wedge moves. Transaction value and platform revenue can separate because of monetization decisions, customer incentives, gross-versus-net revenue recognition, or changes in reporting perimeter. The thesis reconciles the largest divergences against the revenue components issuers themselves disclose, so that movement in the wedge can be attributed either to platform economics or to accounting presentation rather than treated as a fixed structural ratio.

Third, it tests whether the wedge is specific to Indonesia. A separate module of forty-eight matched issuer-years across eight non-Indonesian platform businesses establishes whether the transaction-revenue boundary is a general property of platform intermediation and how far its magnitude varies with business model. This converts the wide cross-platform spread in measured ratios from an obstacle to comparison into a result in its own right.

Fourth, it connects the platform accounting boundary to participant records and tax administration in Indonesia. Statistics Indonesia evidence identifies where e-commerce activity and business recordkeeping occur beyond issuer accounts, while PMK 37/2025 and Directorate General of Taxes materials show how marketplace-held seller identity and transaction-linked turnover enter third-party reporting and withholding.

~~Second, I use Malaysia's January 2024 low-value-goods expansion as a descriptive policy benchmark, reporting both the policy's observed collections and a two-country before-and-after comparison with Vietnam without a causal claim. Third, I rebuild GoTo's 2020–2023 Group series from primary filings. The unbooked share remains between 97.6% and 98.9%, showing a persistent transaction-revenue gap rather than a monotonic increase.~~

Fourth, I use scope-matched historical Grab and Shopee disclosures to test whether large transaction-to-revenue gaps persist within platforms, and a 47-event earnings-announcement panel to test short-window investor response. The event study finds no stable association between the ecosystem ratio and market-adjusted cumulative abnormal returns.

1.2 What this paper does and does not claim1.3 What this proposal does and does not claim

Table 1 states the Table 2 states the paperproposal's inferential boundaries before the estimates are presented. Appendix A records the revision and provenance trail separately so that the main text can focus on the final design and results.

Table 1. Claims supported—and not supported—by the paper's evidenceTable 2. Claims supported and not supported by the evidence assembled for this proposal.

Established here	Not established here
An order-of-magnitude regional estimate built from one-Indonesia-focused three-platform base and four calibrated-rescalings.	Five independent country measurements or a complete-measure of ASEAN's informal economy.
Two descriptive Malaysia benchmarks: about US\$104M of directly-reported LVG collections (RM476M as reported; converted at 2024-period-average FX) and an approximately US\$136M historical FX before-and-after difference.	A causal effect identified by matched tax bases, parallel-trends, or a statistically-informative panel.
A primary-source GoTo Group series in which the unbooked share remains between 97.6% and 98.9% over-2020–2023.	That the GoTo Group series is perfectly country-isolated or that its unbooked share rose monotonically.
The wedge measures transaction value not booked as platform revenue.	That all participant-facing GTV is profit, taxable income, unpaid tax, or GDP value-added.
<u>A longitudinal Indonesian series of the transaction–revenue wedge across seventeen admitted platform-years, reported by evidence tier.</u>	<u>A population estimate of Indonesia's platform economy, or a complete measure of activity outside platform accounts.</u>
<u>An arithmetic reconciliation of selected large divergences against disclosed revenue components.</u>	<u>A causal decomposition of why monetization, incentives, or perimeter changed.</u>
<u>A legal route from marketplace-held seller and transaction records into third-party reporting and withholding.</u>	<u>That compliance, collections, or enforcement outcomes have improved under PMK 37/2025.</u>
The 47-event investor-response panel finds no stable-relationship between the ecosystem ratio and short-window-market-adjusted CAR across the primary specifications.	That the ecosystem ratio is economically irrelevant to-investors in all settings, or that no alternative specification-could detect a relationship.

~~2. Literature review and theoretical framework~~ 2. Literature Review and Research Gap

2.1 Platform economics and the location of value

The economics of multisided platforms begins with the interaction of distinct user groups whose participation affects the value received by the other side. Rochet and Tirole (2003), Caillaud and Jullien (2003), Parker and Van Alstyne (2005), and Armstrong (2006) show why platform pricing cannot be read as a simple markup over cost: prices and commissions are chosen to balance cross-group network effects, participation, and competition. Hagiu and Wright (2015) further distinguish a multisided platform from a vertically integrated firm by locating decision rights and entrepreneurial effort with affiliated participants rather than inside the platform. Evans and Schmalensee (2016) translate the same point into the business-model literature. These results explain why low retained shares can coexist with large ecosystems and why platform revenue is a poor proxy for the transaction value coordinated through the platform.

2.2 Revenue recognition and the accounting boundary

Accounting rules and business-model design determine how transaction value becomes platform revenue. Under IFRS 15, principal-agent treatment affects whether revenue is presented gross or net. Customer incentives, advertising and service income, acquisitions, and changes in reporting perimeter can move recognized revenue without an equivalent change in facilitated commerce. Comparability matters because users of financial statements must understand how similar economic events map into reported numbers; De Franco, Kothari, and Verdi (2011) show that comparability is associated with a stronger analyst information environment. These mechanisms determine how the invisible wedge appears in corporate accounts and how it changes through time.

~~2.2 Informality, platform work, and measurement~~ 2.3 Informality, platform work, and measurement

Traditional informal-economy estimates rely on household surveys, firm surveys, or indirect macroeconomic indicators. Medina and Schneider (2019) estimate a large global shadow economy; La Porta and Shleifer (2014) emphasise the small scale and low productivity of informal firms; and Ulyssea (2018) shows that registration and off-the-books employment are distinct margins whose welfare effects cannot be inferred from formal status alone. The International Labour Organization (2021, 2023) likewise treats platform work and informality as overlapping but non-identical categories.

Platform participants occupy an unusual position in this literature. Their transactions are priced, recorded, and settled through a formal intermediary, but the participant may remain self-employed, unregistered, below filing thresholds, or simply outside an automatic reporting channel. The object measured here is therefore not the shadow economy as conventionally defined. It is a

digitally recorded flow whose administrative treatment resembles self-reported income even though an intermediary already possesses the record.

2.3 Third-party reporting as tax capacity**2.4 Third-party reporting as tax capacity**

A central result in public finance is that tax compliance depends strongly on third-party information and remittance structure. Kleven et al. (2011) find an evasion rate of 0.3 percent on income subject to third-party reporting versus 37 percent on self-reported income in Denmark. Pomeranz (2015) shows that the VAT paper trail creates self-enforcement across firms, while Naritomi (2019) demonstrates that consumer-held records can raise reported sales. Kleven, Kreiner and Saez (2016) formalise firms as fiscal intermediaries, and Slemrod (2019) places information reporting and remittance regimes among the core instruments of modern enforcement.

Jensen (2022) connects this mechanism to development: as employment moves into organisations that keep records and report income, the feasible tax base expands. Platform work creates a new configuration - a large intermediary with detailed records but, in many jurisdictions, no comprehensive obligation to report seller or worker income. Collins et al. (2019) and Garin et al. (2023) show how platform information returns affect the measured prevalence of gig work in the United States. The administrative problem is therefore not a lack of data generation; it is the absence, incompleteness, or fragmentation of the reporting rule.

Two finance studies extend this administrative-data channel. Barrios, Hochberg, and Yi (2022) show that gig-platform entry increases new business formation and small-business lending, while Denes, Lagaras, and Tsoutsoura (2025) use U.S. tax-return microdata to show that gig work can serve as a pathway into entrepreneurship. Together, these studies reinforce the idea that platform-mediated work leaves economically meaningful financial and administrative records.

2.4 Digital taxation and platform-reporting rules**2.5 Digital taxation and platform-reporting rules**

Corporate digital-tax reforms and platform-participant reporting address different bases. The OECD Pillar One blueprint concerns the allocation of taxing rights over large multinational enterprises (OECD 2020b). By contrast, the OECD's Forum on Tax Administration report and Model Rules require platforms to collect and report information on income realised by sellers and service providers (OECD 2019, 2020a). The European Union's DAC7 regime similarly imposes due-diligence and reporting duties on platform operators (European Union 2021). These instruments provide the closest policy precedent for the mechanism studied here: using the platform's existing transaction record to improve participant-level visibility without treating platform revenue as the participant tax base.

Indonesia's PMK 37/2025 applies the same logic through seller identification, transaction-linked reporting, and marketplace withholding.

2.6 Research gap

Taken together, these literatures explain why transaction value can exceed platform revenue, why accounting presentation can move the two apart, why participant reporting remains incomplete, and why third-party information matters for tax administration. What remains less developed is a longitudinal, source-auditable account for a single market that measures the invisible wedge, explains why it changes, and connects the digitally recorded participant-facing flow to the reporting mechanisms through which a tax authority can observe it. This thesis addresses that gap for Indonesia.

3. Theoretical framework: the ecosystem ratio

3.1 The measure

To capture the invisible wedge in Indonesia, I propose and construct a proxy To capture the invisible wedge in Indonesia, I propose and construct a proxy called the Ecosystem Ratio. The measure is my own construction for this paper rather than one adopted from a prior study. It rests on Sections 2.1 and 2.2: booked revenue need not track coordinated transaction value, and the share entering revenue is itself a reporting choice.

For example, if a platform processes 100 units of transaction value and books 10 units as revenue, 90 units pass through the ecosystem without being recorded as the platform's own revenue. The ecosystem ratio formalizes that comparison: $(100 - 10) / 10 = 9$.

$$\text{Ecosystem ratio} = (\text{GTV} - \text{platform revenue}) / \text{platform revenue} \quad (1)$$

Gross transaction value (GTV), or gross merchandise value (GMV) in some platform disclosures, is the total value of transactions processed through a platform, regardless of who ultimately receives the proceeds. This construction is consistent with the platform-economics literature showing that booked platform revenue need not track the total transaction value coordinated through a multisided platform (Rochet and Tirole 2003; Evans and Schmalensee 2016).¹

The ecosystem ratio describes the scale of participant-facing transaction value relative to the platform's auditable revenue base. A higher ratio indicates a larger unbooked transaction flow relative to the platform's own revenue, and therefore a wider administrative information gap; it does not by itself imply greater participant profit, taxable income, value-added, non-compliance, or tax due.

This distinction is essential. Platform filings and investor disclosures can make the accounting gap measurable, but the tax treatment of the residual varies according to participant type, deductible

¹The paper measures gross participant-facing transaction value that is not booked as platform revenue; it does not estimate participant profit, unpaid tax, or value-added.

expenses, tax thresholds, registration status, and taxes already remitted through other channels. The empirical contribution of my paper is to quantify the scale of the recorded transaction flow that is not directly captured by the existing tax system.

3.2 Movement over time

The ecosystem ratio measures scale at a point in time. Because the transaction–revenue relationship also changes, the analysis defines a growth-divergence measure.

$$\text{Growth divergence} = \text{transaction-value growth} - \text{platform-revenue growth} \quad (2)$$

A value of zero indicates that transaction value and revenue move together and the wedge is stable in relative terms. A positive value indicates transaction value growing faster than revenue, widening the wedge; a negative value indicates the reverse. Opposite signs identify periods in which transaction activity and platform revenue move in different directions, which are the cases where a monetization or reporting mechanism is most likely to be doing the work. Material movements are then reconciled against disclosed revenue components: monetization, customer incentives, gross-versus-net revenue recognition, and changes in business scope.

4. Data, methodology, and provenance4. Data and Methodology

4.1 Institutional background: Grab, GoTo, and Shopee4.1 Empirical setting and platform selection

~~I focus on Grab, GoTo, and Shopee (operated by Sea Limited) because they are large ASEAN-origin consumer platforms that directly intermediate transactions and provide public financial disclosures sufficient to observe or transparently derive transaction value and platform revenue.~~Platform selection follows from what can actually be measured. The Ecosystem Ratio requires both sides of the transaction–revenue boundary for the same period and the same business scope. Public listing makes these platforms unusually observable relative to private competitors, which is essential for the company-level measurement used here.

Grab does not disclose gross transaction value by country; its transaction measures are reported by business line, so an Indonesian figure must be derived from a company-wide take rate. Sea Limited discloses revenue for Southeast Asia as a single region with no Indonesia breakout, so Shopee’s Indonesian transaction value must be drawn from an external market-research source. GoTo reports both inputs directly, but only at Group level, which includes operations outside Indonesia. No listed platform therefore publishes a strictly country-labelled matched pair for Indonesia. Three Indonesian issuers come closest and carry the longitudinal evidence: PT Global Digital Niaga Tbk (Blibli), PT Bukalapak.com Tbk, and Tokopedia, reported as an e-commerce segment within PT GoTo Gojek Tokopedia Tbk. Each reports transaction value and revenue for the same period from a single issuer disclosure, but their perimeters differ: Tokopedia’s e-commerce segment is Indonesia-aligned, Blibli’s 3P Retail segment also contains travel, and Bukalapak’s

Group figures include overseas operations. Section 4.2 therefore admits every observation under an explicit evidence tier rather than treating any of them as an Indonesia-only measurement.

~~Grab Holdings Limited. Grab began as a ride-hailing app in Malaysia in 2012 and now operates Mobility, Deliveries, and Financial Services across Southeast Asia. It listed on Nasdaq in December 2021 through a special-purpose acquisition company (SPAC) merger. For this paper, Grab's relevant transaction measure is On-Demand GMV, which combines Mobility and Deliveries; Financial Services is reported separately and is excluded from that GMV measure.~~

~~PT GoTo Gojek Tokopedia Tbk (GoTo). GoTo was formed through the 2021 merger of Gojek and Tokopedia and listed on the Indonesia Stock Exchange in April 2022. Indonesia is its core market, but Group disclosures are not a country-level split. From 2024, Tokopedia's e-commerce operations were deconsolidated after the TikTok Shop transaction, creating a structural break in GoTo's reported GTV and revenue series. Pre- and post-2024 figures are therefore not fully like-for-like.~~

~~Sea Limited / Shopee. Sea Limited, headquartered in Singapore, operates Shopee (e-commerce), Garena (digital entertainment), and Monee, formerly SeaMoney (digital financial services). Sea listed on the NYSE in 2017. Because consolidated Sea revenue includes all three businesses, the analysis pairs Shopee GMV with Shopee e-commerce revenue whenever the filings permit rather than mixing a segment numerator with a consolidated denominator.~~

Scope note. These platforms do not provide interchangeable measurements of one common business model. ~~Grab and GoTo are centered on on-demand services, while Shopee is centered on e-commerce~~Grab is centered on on-demand services, while Shopee, Tokopedia, Bilibli, and Bukalapak are centered on e-commerce; their geographic coverage and revenue-recognition conventions also differ. ~~For that reason, the 2023 base is described as Indonesia-focused, cross-platform ratio levels are interpreted cautiously, and the four non-Indonesian country figures are treated as calibrations rather than direct measurements.~~For that reason cross-platform ratio levels are interpreted cautiously, and the supporting FY2023 cross-section is reported separately from the core longitudinal sample rather than pooled with it.

~~4.2 Sample construction~~4.2 Admission rules and evidence tiers

Platform revenue and transaction value are taken from primary company disclosures: ~~Grab and Sea Limited Form 20-F filings and investor releases, and GoTo annual and quarterly reports.~~ Momentum Works (2024) is used only where an Indonesia-specific market-share input is not separately disclosed. The core cross-platform sample is fiscal year 2023, and Table 2 identifies whether each input is directly reported, derived for Indonesia, or reported only at GoTo Group

level. A platform-year enters the core sample only when both measures cover the same period; the geographic and business scope can be identified; units, currency, and definitions are known; derived inputs can be traced to source disclosures; and material structural breaks are flagged rather than treated as ordinary year-to-year change. Annual totals, quarterly figures, and repeated publication vintages of the same period are not counted as independent observations. All monetary values are reported in U.S. dollars, converted from Indonesian rupiah at period-average official rates, with the native-currency figure retained in the source file.

Take rate is defined as platform revenue divided by gross transaction value (GTV). Table 3 sets out the three tiers, the series admitted to each, and the number of levels and within-series transitions each contributes. The base estimate's blended take rate is the same ratio computed across all three platform rows combined—total platform revenue divided by total GTV, 5.88%—and is used only for the four-country scenario calibration in Section 7. Because the GoTo row is Group-level, I refer to this as the Indonesia-focused blended take rate rather than a perfectly country-isolated Indonesia parameter.

A separate historical sample tests persistence within platforms without mixing geographic scopes. It extends scope-matched Grab Mobility + Deliveries data through 2019-2021 and Shopee GMV and segment revenue through 2017-2021; these observations are reported separately from the Indonesia-focused Table 3.

Table 2. Source, derivation, and geographic scope of each 2023 Indonesia-focused platform input Table 3. Evidence tiers for Indonesian platform observations. No strictly country-labelled matched pair exists, so each observation is admitted under one tier and reported under that label; tiers are never pooled into a single monetary total. Tokopedia is a business segment rather than an explicit country line; Blibli 3P Retail also contains tiket.com travel and Bukalapak reports at Group level with overseas operations; Grab and Shopee each require one side to be derived from a Group monetization rate or an external market estimate.

Platform	Metric	Source	Provenance
Grab	Revenue	Grab 2023 Form 20-F	Directly disclosed at country-level.
Grab	GTV	Derived	Indonesia revenue divided by company-wide take rate (11.24%); country GTV is not independently disclosed.
GoTo-Group	GTV and revenue	GoTo FY2023 annual report	Both inputs directly reported at Group level; the Group also operated outside Indonesia, and no country-level GTV split is disclosed.
Shopee	GTV and revenue	Sea 2023 Form 20-F; Momentum Works 2024	Regional revenue/GMV disclosed; Indonesia GMV estimated from market share, with revenue derived from disclosed take rate (10.06%).
Evidence tier	Series	Levels	Transitions
Direct Indonesia-aligned segment	Tokopedia e-commerce	2	1
Direct issuer, scope-pending	Blibli 3P Retail; Bukalapak Group	9	7
Conditional country reconstruction	Grab Indonesia; Shopee Indonesia	6	4

Platform	Metric	Source	Provenance
All tiers (inventory diagnostic)	5 series	17	12

Only GoTo supplies both inputs independently, but its reported figures are Group-level. Grab and Shopee ratios inherit the take-rate assumptions used to derive one or both Indonesia-specific components. Two candidate observations are excluded under these rules. Tokopedia FY2021 is excluded because its transaction and revenue measures cover different periods, and Bukalapak FY2024 is excluded because transaction value covers nine months while revenue covers twelve. Tokopedia's e-commerce operations were deconsolidated following the TikTok Shop transaction in 2024, which is recorded as a structural break rather than as a change in the wedge.

4.3 Candidate inventory across evidence modules

The study is not a single-fiscal-year design. Table 4 lists the seven retained modules. They are reported separately and never summed into one sample size, because their geographic, regulatory and business-model definitions differ.

Table 4. Retained evidence modules, coverage, and role. Each module has a distinct unit of observation; counts are not additive across rows.

Module	Coverage	Retained scale	Role
Direct Indonesian issuer candidates	FY2019–FY2025	13 period-matched platform/segment-years; 12 with positive revenue denominators	Candidate longitudinal core
Conditional country reconstructions	FY2021–FY2024	6 Grab/Shopee platform-years	Sensitivity evidence only
Indonesia marketplace structure	2022–2025	Platform-level market-share and structure evidence	Coverage, competition, structural breaks
BPS official e-commerce evidence	2020–2024	National indicators; 39 provinces observed in 2023 and 2024	Activity, participation, channels, recordkeeping
ASEAN corroboration	2019–2025	42 country-years from 450 source-vintage rows	Separate-country robustness
Global issuer corroboration	Multi-year issuer histories	48 matched issuer-years, 8 businesses, 40 transitions	Business-model corroboration
Historical quarterly accounting	2017–2022	47 company/segment-quarter observations	Historical disclosure evidence

4.4 Derivation of conditional country values

Grab discloses Indonesia revenue but not Indonesia transaction value, so its country transaction value must be derived; Table 5 sets out that derivation year by year. For each year the conditional construction applies Grab's own Group monetization rate to its disclosed Indonesia revenue: $\text{Group monetization rate} = \text{Group revenue} / \text{Group GMV}$, and $\text{conditional Indonesia GMV} = \text{Indonesia revenue} / \text{Group monetization rate}$. The construction assumes that Grab's Indonesian activity shares the Group revenue-to-GMV relationship: transparent and testable, but not a company disclosure

and not independent evidence of an Indonesia-specific rate. Shopee’s Indonesian transaction value is an external market estimate and its revenue is derived from Sea’s disclosed monetization rate, so both Shopee inputs are conditional. The external estimate is taken from Momentum Works, whose annual Southeast Asian e-commerce report is the only recurring public source disaggregating regional marketplace gross merchandise value by country and platform. It is used because Sea Limited discloses no Indonesian figure, is labelled a third-party estimate wherever it enters a calculation, and is tested in the Section 5.1 sensitivity.

Table 5. Worked derivation of conditional Indonesia transaction value for Grab. Indonesia revenue is directly disclosed and Group revenue and Group GMV are source-reported; every Indonesia transaction value in the final column is derived by applying the Group monetization rate, not disclosed by the company.

<u>Year</u>	<u>Indonesia revenue</u>	<u>Group revenue</u>	<u>Group GMV</u>	<u>Conditional Indonesia GMV (derived)</u>
<u>FY2021</u>	<u>US\$79m</u>	<u>US\$675m</u>	<u>US\$16,061m</u>	<u>US\$1,879.7m</u>
<u>FY2022</u>	<u>US\$275m</u>	<u>US\$1,433m</u>	<u>US\$19,937m</u>	<u>US\$3,826.0m</u>
<u>FY2023</u>	<u>US\$605m</u>	<u>US\$2,359m</u>	<u>US\$20,983m</u>	<u>US\$5,381.4m</u>

4.5 Key variables

For each admissible platform-year the analysis records transaction value V, platform revenue R, the absolute wedge $W = V - R$, and the Ecosystem Ratio $E = (V - R) / R$. For each within-series annual transition it records transaction-value growth, revenue growth, and their difference D. Take rate is platform revenue divided by gross transaction value. Material divergences are reconciled against disclosed revenue components — customer incentives, gross and net revenue, monetization, and business scope — where the issuer reports them separately. That reconciliation is arithmetic against the issuer’s own categories; it is not a causal decomposition.

4.6 Evidence beyond platform accounts

BPS-Statistics Indonesia evidence, reported in Section 5.5, locates participant-facing digital commerce outside platform accounts. Bank Indonesia payment-system statistics provide supporting evidence on the infrastructure carrying that commerce; these series measure payment activity, not e-commerce sales. PMK 37/2025 provides for designated marketplaces to identify covered domestic sellers and to withhold and report Article 22 income tax at 0.5 percent of invoice gross turnover, using transaction-linked merchant turnover rather than marketplace corporate revenue. Blibli, Shopee Indonesia, Tokopedia, and Lazada were designated on 1 July 2026, with collection taking effect on 1 August 2026 after a month allowed for system adjustment. Application was subsequently postponed through 31 October 2026, amounts already withheld were ordered returned to sellers, and the mechanism is now scheduled to take effect on 1 November 2026. The Directorate General of Taxes describes the measure as a change in collection mechanism rather than a new tax, so it cannot be read as evidence that sellers previously underpaid. It establishes a

legal reporting architecture; no completed post-implementation period yet exists from which to estimate a compliance effect.

5. Preliminary Evidence and Feasibility

The retained evidence already supports each step of the design: the wedge can be measured, it moves, its movement can be traced to disclosed revenue components, the pattern recurs outside Indonesia, and official statistics locate activity beyond platform accounts. These are feasibility diagnostics for the proposal examination, not final estimates or causal findings, and the final sample admission remains subject to advisor approval.

5.1 FY2023 cross-section

FY2023 is retained as the cleanest common cross-platform snapshot, not as the period of analysis. It is the year for which a detailed Indonesian marketplace-share denominator is available for all three leading platforms. Table 6 reports the wedge and the Ecosystem Ratio for each case and for the three combined.

Table 6. Indonesia-focused FY2023 cross-section. Values in US\$ billions; V is transaction value, R is platform revenue, E is the Ecosystem Ratio. Evidence class: Grab’s transaction value is derived from its Group monetization rate; Tokopedia is a directly reported Indonesia-aligned segment pair; both Shopee inputs combine an external market estimate with a disclosed company-wide rate. The final row sums three documented cases under mixed evidence classes and is not an Indonesia-wide total.

<u>Case</u>	<u>V</u>	<u>R</u>	<u>Wedge</u>	<u>E</u>	<u>Evidence class</u>
<u>Grab Indonesia</u>	<u>5.381</u>	<u>0.605</u>	<u>4.776</u>	<u>7.895×</u>	<u>Derived V</u>
<u>Tokopedia e-commerce</u>	<u>16.331</u>	<u>0.405</u>	<u>15.926</u>	<u>39.296×</u>	<u>Direct pair</u>
<u>Shopee Indonesia</u>	<u>21.520</u>	<u>2.152</u>	<u>19.368</u>	<u>9.000×</u>	<u>Derived V and R</u>
<u>Selected platforms</u>	<u>43.233</u>	<u>3.162</u>	<u>40.070</u>	<u>12.671×</u>	<u>Sum, not a total</u>

The selected-platform wedge of US\$40.07 billion is equivalent to approximately 2.9 percent of Indonesia’s 2023 nominal gross domestic product. Because two of the three cases depend on an assumption, every assumed parameter is varied one at a time and the result is also reported with each platform excluded in turn. Varying Grab’s implied Indonesia monetization rate moves the selected-platform wedge between US\$39.17 billion and US\$41.42 billion; varying Shopee’s assumed Indonesian market share from 35 to 45 percent gives US\$37.65 billion to US\$42.49 billion; varying Shopee’s revenue-to-GMV rate gives US\$39.86 billion to US\$40.29 billion; and varying the Indonesian e-commerce market denominator gives US\$38.13 billion to US\$42.01 billion. Excluding Grab leaves a wedge of US\$35.29 billion across the two remaining cases, excluding Tokopedia leaves US\$24.14 billion, and excluding Shopee leaves US\$20.70 billion. No single assumption or platform drives the order of magnitude. Reproduction checks on the sample,

the applied rate and the aggregate are retained in the data package. These checks do not convert three documented cases into a statistical sample.

5.2 Longitudinal divergence under two evidence universes

Two valid longitudinal summaries coexist because they admit different evidence. Both are reported. Table 7 reports both. The all-tier inventory includes the conditional Grab and Shopee reconstructions; the direct-candidate sensitivity excludes them and instead extends Blibli with its directly transcribed FY2020 prospectus observation.

Table 7. Within-series annual growth divergence under two admission rules. The counts differ because evidence admission differs; the difference is reported rather than resolved into a single headline.

	<u>All-tier inventory</u>	<u>Direct-candidate sensitivity</u>
<u>Series admitted</u>	<u>Blibli FY2021–25, Bukalapak, Tokopedia, Grab, Shopee</u>	<u>Blibli FY2020–25, Bukalapak, Tokopedia</u>
<u>Transitions</u>	<u>12</u>	<u>9</u>
<u>Revenue grows faster</u>	<u>10</u>	<u>6</u>
<u>Transaction grows faster</u>	<u>2</u>	<u>3</u>
<u>Sign reversals</u>	<u>2</u>	<u>3</u>
<u>Median absolute divergence</u>	<u>42.02 pp</u>	<u>42.94 pp</u>

The additional sign reversal in the direct-candidate universe is Blibli FY2020 → FY2021. That the reversal count changes with admission is itself informative: it demonstrates on the study’s own data the proposition that measurement choices can alter an empirical conclusion, which is why the sample hierarchy is shown rather than collapsed.

5.3 Mechanism: what moves the wedge

Tokopedia FY2022–FY2023 is the strongest direct case. Transaction value falls 8.90 percent while third-party net segment revenue rises 53.20 percent. Reconciling the reported revenue components, approximately 60.56 percent of the increase in net revenue is associated with lower customer incentives and 39.44 percent with higher gross revenue; the disclosed components sum exactly to reported net revenue in both years. Recognized revenue therefore strengthened while facilitated activity weakened, for reasons visible in the issuer’s own disclosure. A second, weaker case points the same way. Blibli 3P Retail FY2022–FY2023 shows transaction value rising 34.72 percent while segment net revenue rises 465.33 percent. That divergence is arithmetically real but arises from a very low base — segment net revenue moves from IDR 199 billion to IDR 1,125 billion, about US\$13 million to US\$74 million — and the segment perimeter also contains travel, so it is reported as evidence of a monetization-regime change rather than as a measurement of the same quantity as the Tokopedia case.

5.4 Is the wedge specific to Indonesia?

A separate module tests whether the same measurement boundary appears in platform businesses outside Indonesia. Across 48 matched issuer-years for eight businesses — eBay, Etsy, Shopify,

Jumia, Zalando, Rakuten, Mercado Libre, and Sea — the transaction-to-revenue relationship is present throughout, but its level is business-model determined: take rates range from 0.22 percent to 74.63 percent. Of 40 annual transitions, 29 are clean-scope, with a median absolute growth divergence of 7.10 percentage points and four transitions in which transaction value and revenue move in opposite directions. The wedge is therefore a general property of platform intermediation rather than an Indonesian artefact, while its magnitude is not a stable parameter across business models. Issuers without a continuous matched transaction series — Amazon, Alibaba, PDD, and Coupang — are excluded and the reason recorded.

5.5 Official statistics and institutional evidence

BPS evidence locates digital commercial activity beyond platform accounts. National e-commerce value rises from Rp1,100.87 trillion in 2023 to Rp1,288.93 trillion in 2024, approximately US\$72.2 billion to US\$81.4 billion, an increase of 17.08 percent, while estimated e-commerce businesses rise from 3,816,750 to 4,400,972, an increase of 15.31 percent; implied nominal value per estimated business therefore rises only 1.54 percent. The channel split is more striking: marketplace value rises 1.42 percent against 20.58 percent for the non-marketplace component, so approximately 98.49 percent of the nominal increase falls outside the marketplace component that platform accounts observe. National financial-report ownership among e-commerce businesses is 17.15 percent in 2024. BPS also publishes an analysis reporting higher financial-report ownership among marketplace users than non-users; that is BPS's own published result and is cited as such. Province-level evidence is ecological and is not used to infer a business-level relationship.

Taken together, the modules show that the wedge can be measured under explicit admission rules, that it moves for identifiable reasons, that the pattern is not specific to Indonesia, that most recent national growth occurs outside the channel platform accounts observe, and that a legal route now exists from marketplace records into tax administration. The research question can be answered with evidence that is available, tiered, and auditable to source.

6. Limitations and Advisor Decisions

The principal limitation is that no Indonesian platform publishes a strictly country-labelled matched pair of transaction value and revenue. Every observation is therefore admitted under a named evidence tier, and the tiers differ in what they can support. Tokopedia is an Indonesia-aligned business segment rather than a country line. Blibli's 3P Retail perimeter includes online travel. Bukalapak's Group figures include overseas operations. Grab's Indonesian transaction value is derived from a Group monetization rate, and both Shopee inputs combine an external market estimate with a disclosed company-wide rate. These are properties of what issuers disclose, not choices made for convenience, and they bound what the study can claim.

Three further boundaries apply. Mechanism reconciliations are arithmetic against the issuer’s own disclosed categories, not causal decompositions. BPS province evidence is ecological and cannot identify a business-level relationship. PMK 37/2025, postponed to 1 November 2026, offers no completed period from which to estimate a compliance or revenue effect.

Two decisions are sought at the proposal examination. First, whether the direct issuer scope-pending tier — Blibli 3P Retail and Bukalapak Group — should be admitted to the final longitudinal main sample, reported only as a labelled tier, or excluded. Second, whether the conditional country reconstructions for Grab and Shopee may appear in the main comparison or should be confined to sensitivity analysis. Both decisions change the reported transition counts, as Table 7 shows, and neither can be settled by the data alone.

7. Work Plan

Table 8 sets out the remaining work. The empirical backend is assembled and source-linked, so the outstanding tasks are sample admission, any analysis the committee requests, and construction of the final manuscript under the approved boundary.

Appendix A. Definition of Variables and Data Sources

Table A1. Variables used in the analysis, their definitions, and the source of each. Transaction value and platform revenue are observed from issuer disclosures; the wedge, the Ecosystem Ratio, the growth rates and their difference are derived from unrounded source inputs.

Variable	Definition	Source
V	Gross transaction value processed through the platform in the period, at the stated business and geographic scope	Issuer annual reports, prospectuses and results materials
R	Platform revenue recognized for the same period and scope	Issuer annual reports and financial statements
$W = V - R$	Absolute invisible wedge: transaction value not booked as platform revenue	Derived
$E = (V - R) / R$	Ecosystem Ratio: the wedge relative to booked platform revenue	Derived
$g(V), g(R)$	Year-on-year growth in transaction value and in platform revenue	Derived
$D = g(V) - g(R)$	Growth divergence between transaction value and platform revenue	Derived
Take rate	Platform revenue divided by gross transaction value	Derived

<u>Variable</u>	<u>Definition</u>	<u>Source</u>
<u>Revenue components</u>	<u>Customer incentives, third-party gross revenue, third-party net revenue</u>	<u>Issuer segment notes where separately disclosed</u>
<u>Evidence tier</u>	<u>Direct Indonesia-aligned, direct issuer scope-pending, or conditional country reconstruction</u>	<u>Assigned by the admission rules in Section 4.2</u>
<u>E-commerce value</u>	<u>National nominal e-commerce transaction value</u>	<u>BPS-Statistics Indonesia, E-Commerce Statistics</u>
<u>E-commerce businesses</u>	<u>Estimated number of businesses conducting e-commerce</u>	<u>BPS-Statistics Indonesia, E-Commerce Statistics</u>
<u>Financial-report ownership</u>	<u>Share of e-commerce businesses holding financial statements</u>	<u>BPS-Statistics Indonesia, national indicators</u>

Table 8. Planned work to the final defence. The empirical backend is assembled and source-linked; the remaining work is admission, requested analysis, and manuscript construction under the approved boundary.

<u>Period</u>	<u>Planned work</u>
<u>September 2026</u>	<u>Advisor review of the data package; proposal oral examination; freeze the final sample-admission rule and the checks the committee requests.</u>
<u>October–November 2026</u>	<u>Complete requested mechanism reconciliations and source concordance; extend the sensitivity analysis to the approved tier boundary; incorporate committee feedback; freeze thesis tables and figures.</u>
<u>Before final defence</u>	<u>Rebuild the full manuscript under the approved sample; retain the global, official-statistics and institutional modules only where they advance the final argument.</u>

References

- Armstrong, M. (2006). Competition in two-sided markets. *RAND Journal of Economics*, 37(3), 668–691.
- Barrios, J. M., Hochberg, Y. V., & Yi, H. (2022). Launching with a parachute: The gig economy and new business formation. *Journal of Financial Economics*, 144(1), 22–43.
<https://doi.org/10.1016/j.jfineco.2021.12.011>
- Benhassine, N., McKenzie, D., Pouliquen, V., & Santini, M. (2018). Does inducing informal firms to formalize make sense? Experimental evidence from Benin. *Journal of Public Economics*, 157, 1–14.
<https://doi.org/10.1016/j.jpubeco.2017.11.004>
- Brown, S. J., & Warner, J. B. (1985). Using daily stock returns: The case of event studies. *Journal of Financial Economics*, 14(1), 3–31. [https://doi.org/10.1016/0304-405X\(85\)90042-X](https://doi.org/10.1016/0304-405X(85)90042-X)
- Caillaud, B., & Jullien, B. (2003). Chicken & egg: Competition among intermediation service providers. *RAND Journal of Economics*, 34(2), 309–328.
- Collins, B., Garin, A., Jackson, E., Koustas, D., & Payne, M. (2019). Is gig work replacing traditional employment? Evidence from two decades of tax returns. U.S. Internal Revenue Service, Statistics of Income Working Paper.
- de Mel, S., McKenzie, D., & Woodruff, C. (2013). The demand for, and consequences of, formalization among informal firms in Sri Lanka. *American Economic Journal: Applied Economics*, 5(2), 122–150.
<https://doi.org/10.1257/app.5.2.122>

- Denes, M. R., Lagaras, S., & Tsoutsoura, M. (2025). Entrepreneurship and the gig economy: Evidence from U.S. tax returns. *Journal of Financial Economics*, 173, 104156. <https://doi.org/10.1016/j.jfineco.2025.104156>
- Directorate General of Taxes Indonesia. (2025). Pajak atas Usaha Ekonomi Digital Terkini: Rp32,32 Triliun. Press release, 20 January.
- European Union. (2021). Council Directive (EU) 2021/514 amending Directive 2011/16/EU on administrative cooperation in the field of taxation (DAC7).
- Evans, D. S., & Schmalensee, R. (2016). *Matchmakers: The New Economics of Multisided Platforms*. Harvard Business Review Press.
- Garin, A., Jackson, E., Koustas, D. K., & Miller, A. (2023). The evolution of platform gig work, 2012–2021. NBER Working Paper No. 31273. <https://doi.org/10.3386/w31273>
- General Department of Taxation Vietnam. (2024). Renewing electronic tax administration delivers effective budget-collection results. 27 December.
- Google, Temasek, and Bain & Company. (2024). e-Conomy SEA 2024: Profits on the Rise, Harnessing SEA's Advantage.
- GoTo Gojek Tokopedia Tbk. (2022). GoTo Group Reports Second Quarter and First Half 2022 Financial Results.
- GoTo Gojek Tokopedia Tbk. (2023). Annual Report, Financial Year 2022. Indonesia Stock Exchange.
- GoTo Gojek Tokopedia Tbk. (2024). Annual Report, Financial Year 2023. Indonesia Stock Exchange.
- Grab Holdings Limited. (2022). Form 20-F, Fiscal Year 2021. U.S. Securities and Exchange Commission.
- Grab Holdings Limited. (2024). Form 20-F, Fiscal Year 2023. U.S. Securities and Exchange Commission.
- Hagiu, A., & Wright, J. (2015). Multi-sided platforms. *International Journal of Industrial Organization*, 43, 162–174. <https://doi.org/10.1016/j.ijindorg.2015.03.003>
- International Labour Organization. (2021). *World Employment and Social Outlook 2021: The Role of Digital Labour Platforms in Transforming the World of Work*. ILO.
- International Labour Organization. (2023). Resolution Concerning Statistics on the Informal Economy. 21st International Conference of Labour Statisticians.
- Jensen, A. (2022). Employment structure and the rise of the modern tax system. *American Economic Review*, 112(1), 213–234.
- Kleven, H. J., Knudsen, M. B., Kreiner, C. T., Pedersen, S., & Saez, E. (2011). Unwilling or unable to cheat? Evidence from a tax audit experiment in Denmark. *Econometrica*, 79(3), 651–692.
- Kleven, H. J., Kreiner, C. T., & Saez, E. (2016). Why can modern governments tax so much? An agency model of firms as fiscal intermediaries. *Economica*, 83(330), 219–246. <https://doi.org/10.1111/ecca.12182>
- La Porta, R., & Shleifer, A. (2014). Informality and development. *Journal of Economic Perspectives*, 28(3), 109–126.
- ~~MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13–39.~~
- Medina, L., & Schneider, F. (2019). Shedding light on the shadow economy: A global database. CESifo Working Paper 7981.
- ~~Ministry of Finance Malaysia. (2023). Sales tax on imported low-value goods sold online. Press release, 18 December.~~
- Momentum Works. (2024). *Ecommerce in Southeast Asia 2023*.
- Naritomi, J. (2019). Consumers as tax auditors. *American Economic Review*, 109(9), 3031–3072.
- OECD. (2019). *The Sharing and Gig Economy: Effective Taxation of Platform Sellers*. OECD Publishing. <https://doi.org/10.1787/574b61f8-en>
- OECD. (2020a). *Model Rules for Reporting by Platform Operators with respect to Sellers in the Sharing and Gig Economy*. OECD Publishing. <https://doi.org/10.1787/d7973047-en>
- OECD. (2020b). *Tax Challenges Arising from Digitalisation – Report on Pillar One Blueprint*. OECD/G20 Base Erosion and Profit Shifting Project.

- Parker, G. G., & Van Alstyne, M. W. (2005). Two-sided network effects: A theory of information product design. *Management Science*, 51(10), 1494–1504. <https://doi.org/10.1287/mnsc.1050.0400>
- ~~Parliament of Malaysia. (2025). Dewan Negara Hansard, 9 December 2025.~~
- Pomeranz, D. (2015). No taxation without information: Deterrence and self-enforcement in the value added tax. *American Economic Review*, 105(8), 2539–2569.
- Rochet, J.-C., & Tirole, J. (2003). Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4), 990–1029.
- ~~Royal Malaysian Customs Department. (2023). Guidelines for the Implementation of Sales Tax on Low-Value Goods.~~
- Sea Limited. (2019). Form 20-F, Fiscal Year 2018. U.S. Securities and Exchange Commission.
- Sea Limited. (2020). Form 20-F, Fiscal Year 2019. U.S. Securities and Exchange Commission.
- Sea Limited. (2021). Form 20-F, Fiscal Year 2020. U.S. Securities and Exchange Commission.
- Sea Limited. (2022). Form 20-F, Fiscal Year 2021. U.S. Securities and Exchange Commission.
- Sea Limited. (2024). Form 20-F, Fiscal Year 2023. U.S. Securities and Exchange Commission.
- Slemrod, J. (2019). Tax compliance and enforcement. *Journal of Economic Literature*, 57(4), 904–954. <https://doi.org/10.1257/jel.20181437>
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review*, 108(8), 2015–2047. <https://doi.org/10.1257/aer.20141745>
- ~~VietnamPlus. (2023). Foreign suppliers paid more than VND8 trillion in taxes in 2023. Vietnam News-Agency, 21 December.~~
- ~~VietnamPlus. (2024). Foreign suppliers pay nearly 8.7 trillion VND in taxes this year. Vietnam News-Agency, 30 December.~~
- ~~World Bank. (n.d.). World Development Indicators: Official exchange rate (LCU per US\$, period-average); indicator PA.NUS.FCRF. Source: International Monetary Fund, International Financial Statistics.~~
- De Franco, G., Kothari, S. P., & Verdi, R. S. (2011). The benefits of financial statement comparability. *Journal of Accounting Research*, 49(4), 895–931.